



Data for Agentic AI



Data is the foundation for building AI agents

- Quality and trustworthy data is what makes any model work and provide meaningful results
- When a project or job is made of thousands of data pipelines and data from multiple sources, observability to the data work flows is critical
- Databand provides data observability



Databand helps to

- Detect and resolve data quality incidents in real-time
 - Reduce business impacts of bad data quality
 - Automate detection & root cause analysis
 - Improve data engineering efficiency & deliver

Solution component

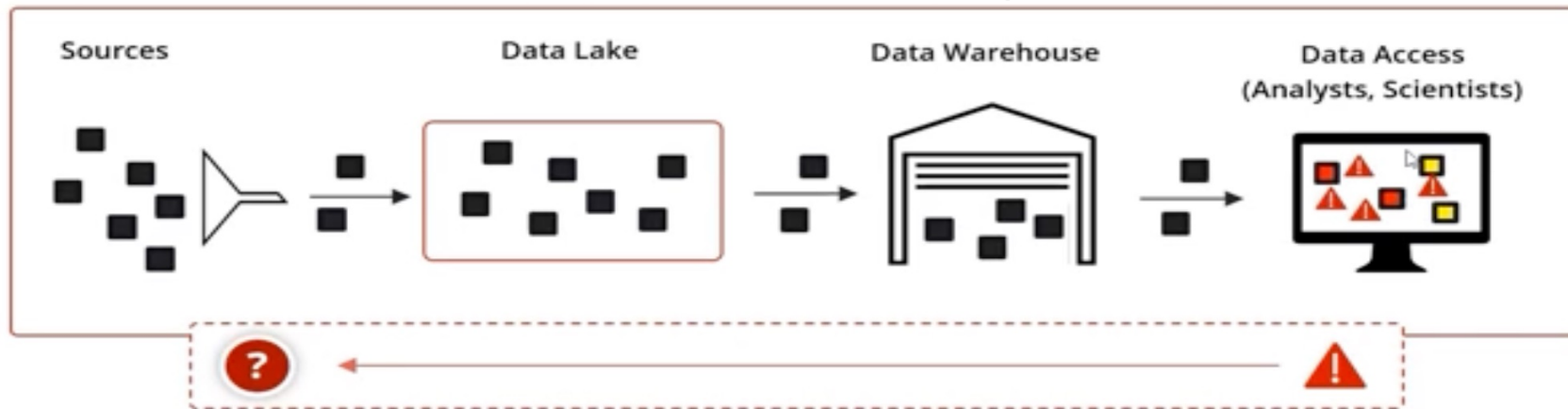
IBM® watsonx.data® integration

IBM® Databand®

<https://www.ibm.com/case-studies/ibm-chief-data-office-databand>

Problem Statement

What happens when we are **reactive** to data issues



Many data quality issues are **overlooked**



Platform only learns about issues when **reported by data consumers**

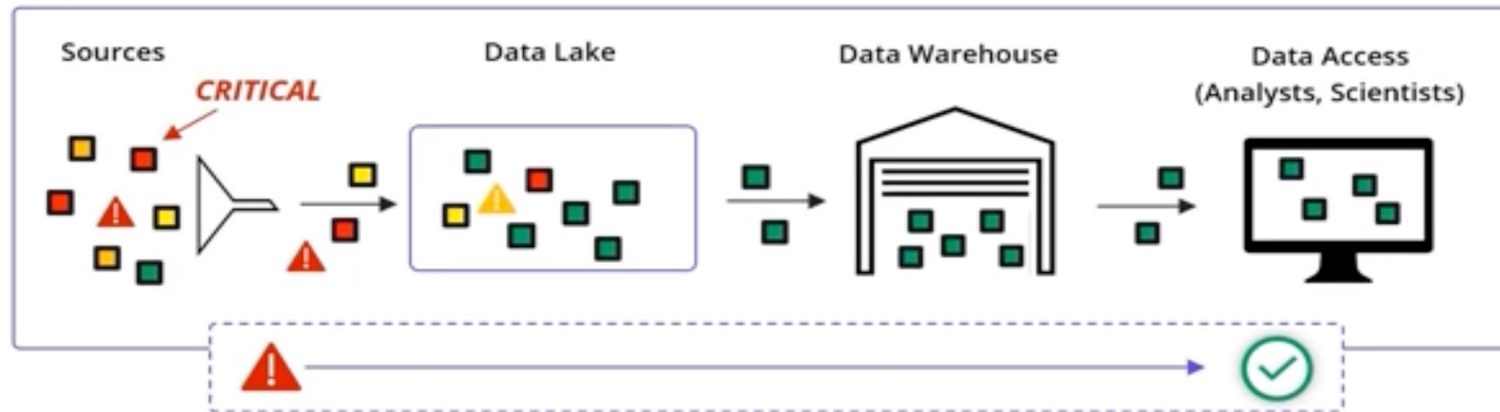


After issues are reported, they **are not resolved quickly**



Solution

Proactive Data Observability: *Shift left* and solve problems at the source



Observe data pipeline **process quality**

- Status
- Performance
- Latency

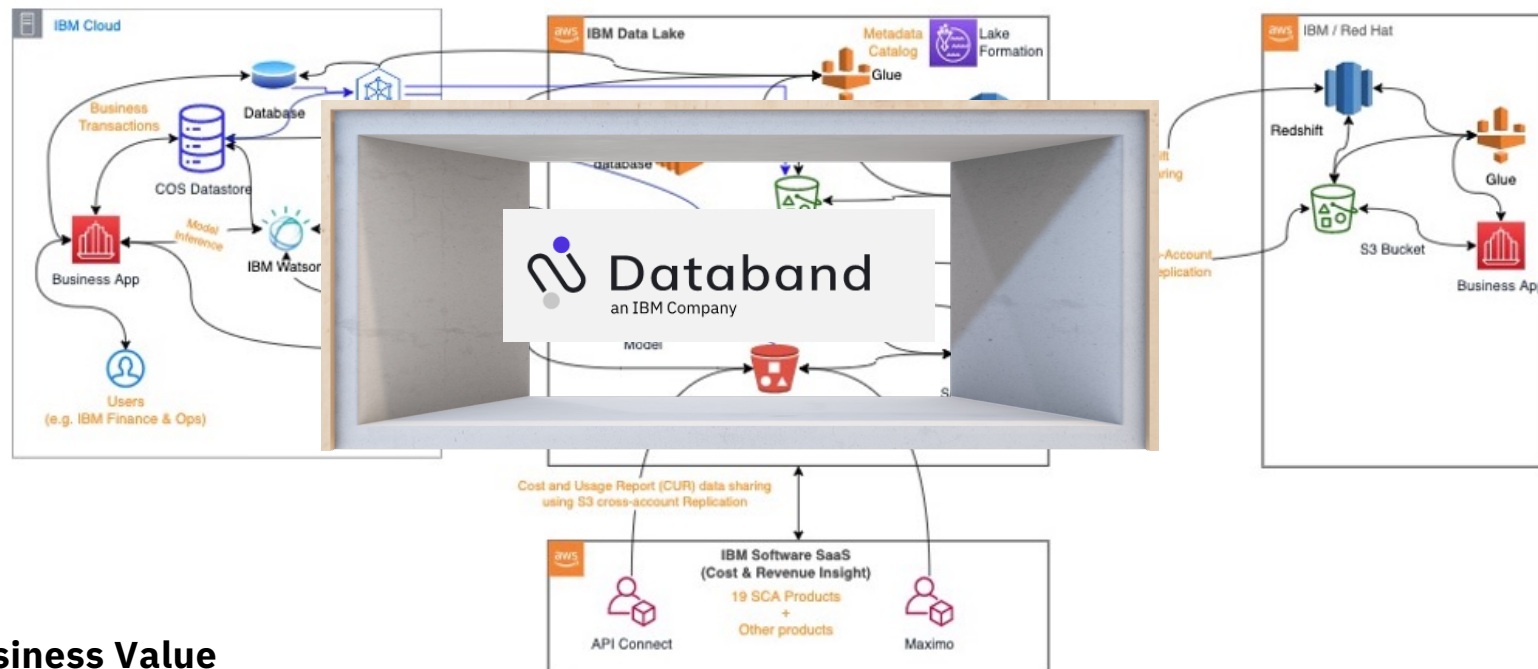
Observe **data quality and reliability**

- *Schema changes*
- *Data shape*
- *Data freshness*

Supported **data pipelines** and **workflow managers**

- *Airflow*: Python, Spark, dbt, SQL and other operators

Databand GCDO M&A Use Cases



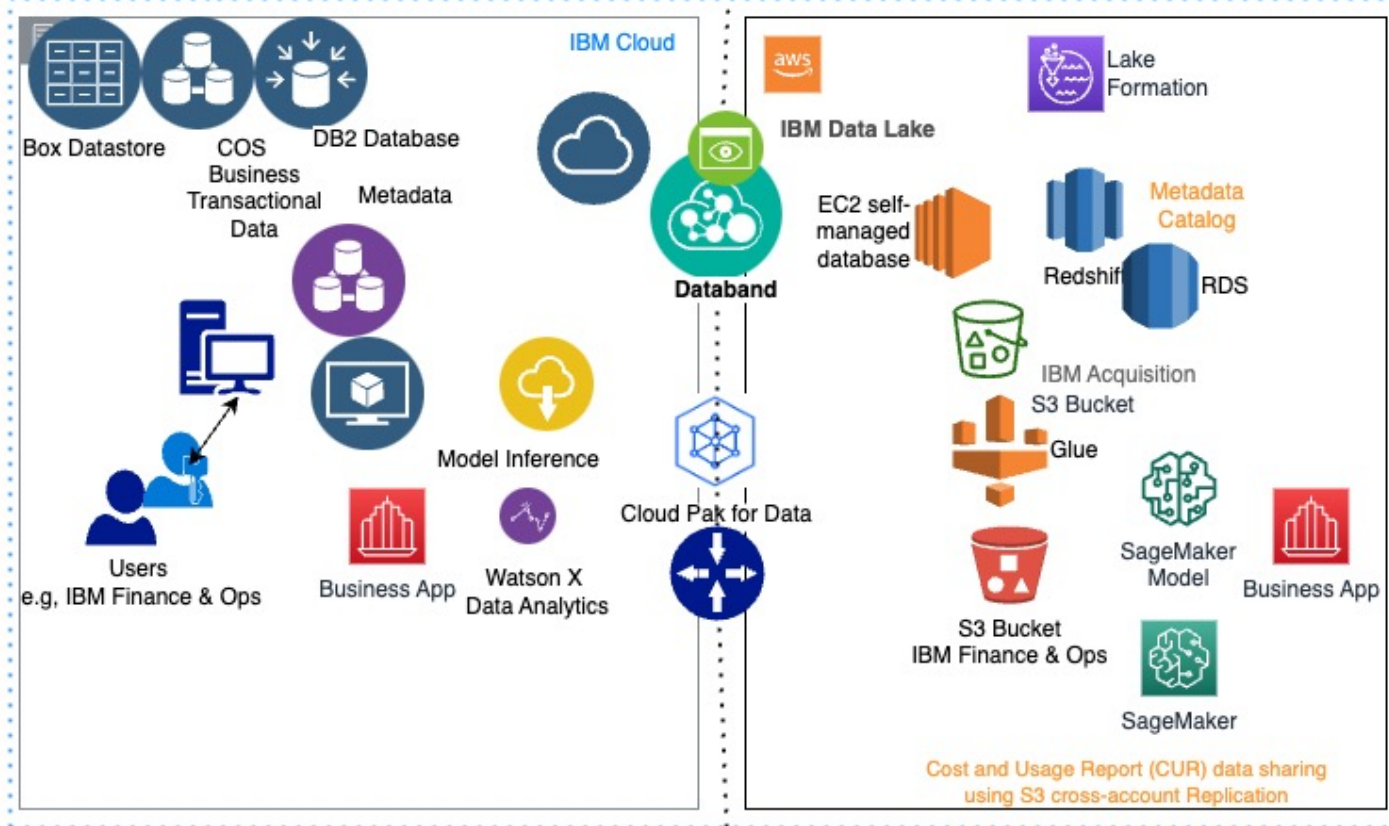
Business Value

Data Observability refers to dynamic monitoring, reporting, and resolving of data incidents at run time. It also provides real time monitoring on data transformation and reports on data quality too. DataBand is an IBM acquired company and the DataBand platform provides autonomous monitoring and reporting (dashboard) and recommends ways to avert / resolve data incident issues. This greatly reduces logging and manual observation / tracking of incidents. The autonomous functions of DataBand enable to realize better SLAs with customers as well as ensure better quality and trust of the data too.

Data reside on multiple clouds, multiple storage mediums

Architecture

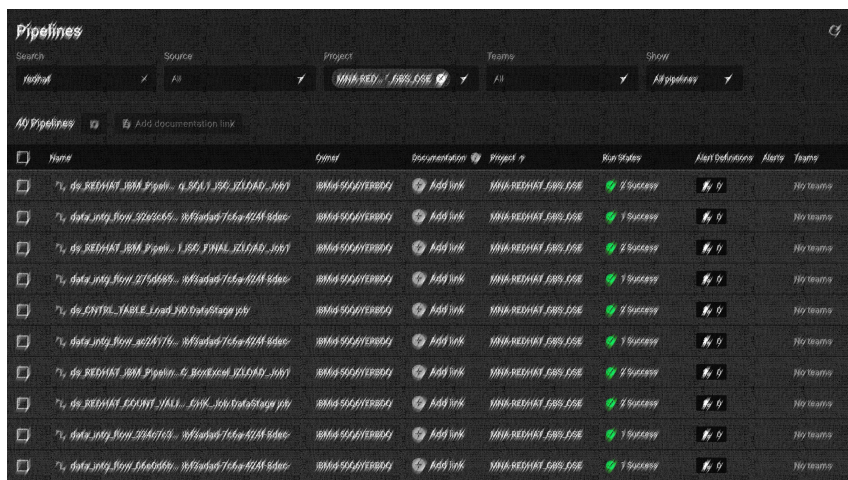
Data Observability with Databand



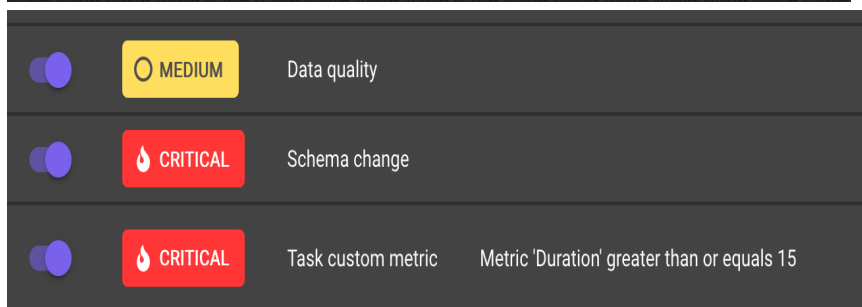
Data Observability with Databand

Apr 26, 2023

- Data Observability with Databand
Pipelines, Job runs and Alerts help focus on those that matters and less resolution time in the event of an issue



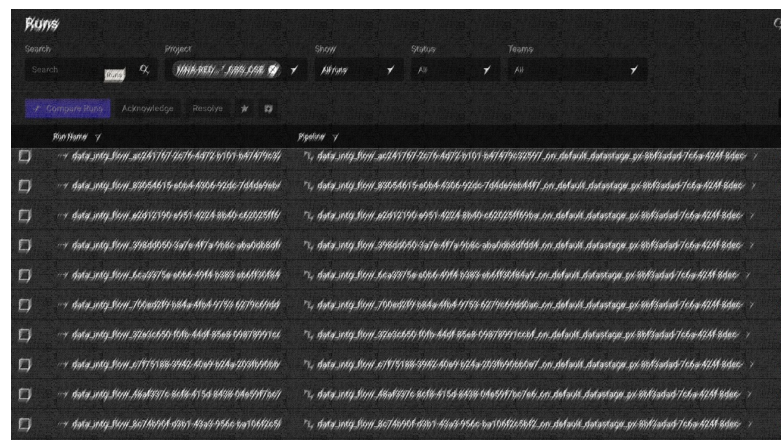
Name	Owner	Documentation	Project	Run Status	Alert Definitions	Alerts	Teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams
data_integ_flow_326545..._ibmdata7cdp42M sdes	ibmdata7cdp42M sdes	Add link	MNA-REDHAT_OBS_OSE	1 Success	0	0	No teams



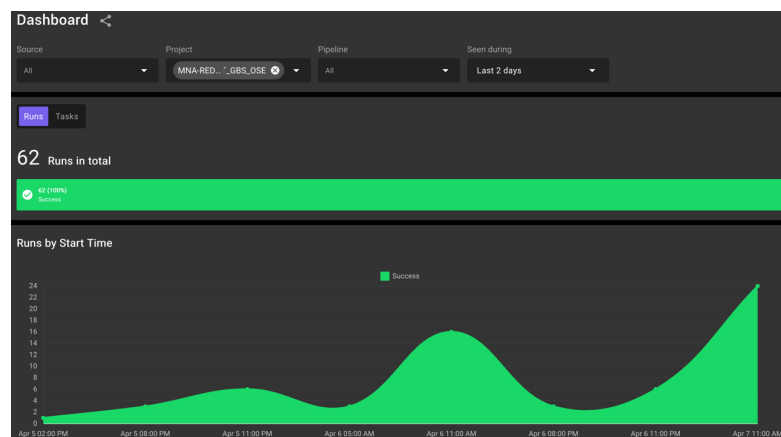
Data quality (MEDIUM)

Schema change (CRITICAL)

Task custom metric (CRITICAL) Metric 'Duration' greater than or equals 15

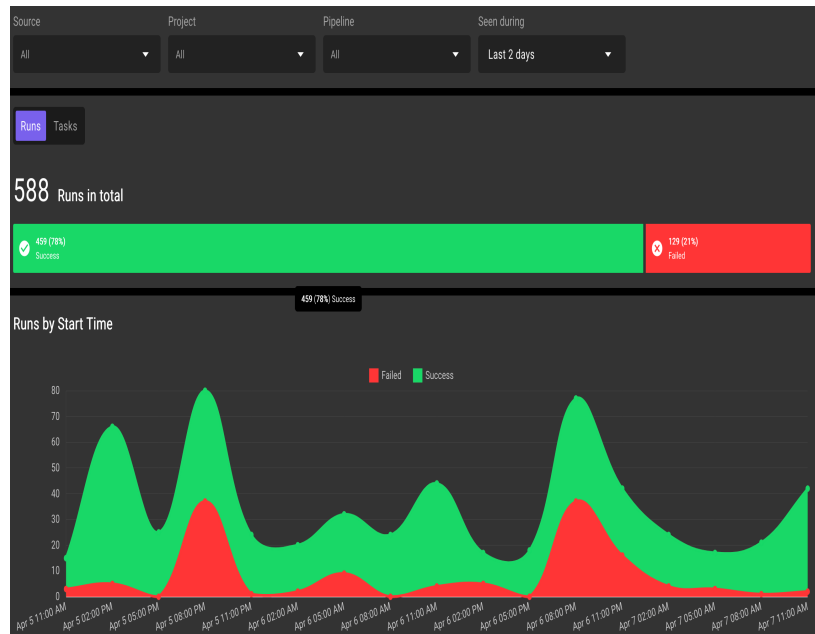


Run Name	Pipeline	Status
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success
data_integ_flow_326545..._ibmdata7cdp42M sdes	data_integ_flow_326545..._ibmdata7cdp42M sdes	Success



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Run Name	Pipeline	Status	Start Time	End Time
ds_UCML_DEV_TRY.DataStage job_8828c52b-b509-4a	ds_UCML_DEV_TRY.DataStage job	Failed	Yesterday at 06:14:38 AM	Yesterday at 06:15:35 AM
ds_UCML_DEV_TRY.DataStage job_0d171c42-4584-4a	ds_UCML_DEV_TRY.DataStage job	Failed	Yesterday at 11:06:36 PM	Yesterday at 11:08:01 PM

Details | Metrics | Logs | Code | Run Info | Data Interactions | Histograms

Running/Failed Tasks

Amazon_S3_1

Error logs

```
07:51 ERROR <Amazon_S3_1,1> at com.ibm.iis.cc.cloud.CloudStagesCloudParallelWriter
07:51 ERROR <Amazon_S3_1,1> at java.lang.Thread.run ()
07:51 ERROR <Amazon_S3_1,1> The write operation was not completed because an excep
07:51 ERROR <Amazon_S3_1,0> The runLocally() of the operator failed.
07:51 ERROR <Amazon_S3_1,1> Operator terminated abnormally: runLocally() did not r
07:51 ERROR <Amazon_S3_1,1> The runLocally() of the operator failed.
07:51 ERROR <Amazon_S3_1,1> Operator terminated abnormally: runLocally() did not r
```

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Tasks

Children | Upstreams | Downstreams | All Tasks | Recursive

- IBM_Db2_on_Cloud_1
- Amazon_S3_1

ds_UCML_DEV...4-3f36c30a09ea

1 m, 25 s

1 m, 1 s | 56 s

IBM_Db2_on_Cloud_1 | Amazon_S3_1

Data Lineage

